Exam Date & Time: 30-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

V SEMESTER B.TECH END SEMESTER EXAMINATION, NOVEMBER 2024

ARTIFICIAL NEURAL NETWORKS [BME 3124]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions.

Instructions to Candidates:
[i] Missing data may be suitably assumed
[ii] No formula book will be provided

1A) Using the gradient descent algorithm, design a binary classifier for a dataset with two features x_1 and x_2, where the initial weights and bias are set to small random values, and the learning rate is 0.05. Describe the steps required to train this classifier, including how the weights and bias are updated through gradient descent. (5)

1B) Given a neural network's output logits for three classes as [2.0, 1.0, 0.1], use the Softmax function to convert these into probabilities. Interpret how these probabilities reflect the network's confidence in its predictions and discuss the impact of one high logit on the others in a multi-class setting. (3)

1C) A feedforward neural network with ReLU in hidden layers and Softmax in the output is trained on a dataset, achieving a validation accuracy of 95%. If the ReLU is replaced by a sigmoid function in all layers, explain how this might impact the training convergence rate. (2)

2A) Using the function $g(x, y, z) = x^2y - y^2z + z^2x$

i. Compute the partial derivatives with respect to x, y, and z.

ii. Formulate the gradient vector and evaluate it at the point (1, 2, 3). (5)

Explain the significance of each partial derivative in determining the direction and rate of change for gradient descent.

2B) After 20 epochs, a neural network reaches a training accuracy of 98%, validation accuracy of 90%, and test accuracy of 88%. Calculate the gap between training and validation accuracy and analyze the implications for overfitting. Analyse and speculate what additional steps could quantitatively bring validation accuracy closer to training accuracy. (3)

2C) Explain why Hopfield networks use binary states of -1 and +1 instead of 0 and 1. How does this choice affect the retrieval process? (2)

3A) Apply the concept of Hebbian learning in a Hopfield network to explain how a pattern is stored in the weight matrix. Given a pattern vector x = [1, -1, 1, -1], compute the weight matrix using the outer product. (3)

3B) Analyze the effect of using a small versus large window size when creating input-output pairs for a sine wave prediction model with an RNN. Describe how each choice impacts the model's ability to capture long-term dependencies and prediction accuracy. (3)

3C) When using an LSTM for sine wave prediction, there are three gates namely forget, input, and output which control information flow. Analyse and explain how each gate contributes to accurately predicting sequential data. (4)

4A) Given an input image of size 32x32x3, a convolutional layer with 8 filters of size 5x5, stride 1, and padding of 2 is applied.

i.Calculate the size of the output feature maps. (4)

ii. Determine the number of feature maps produced.

4B) Explain how the hierarchical structure of CNN layers contributes to feature extraction. Specifically, analyze the types of features captured in the early versus deeper layers. (3)

4C) You are training a CNN on the CIFAR-10 dataset and observe that validation accuracy plateaus while training accuracy continues to improve.

i.Identify the issue indicated by this trend. (3)

ii. Suggest two methods to address this issue.

5A) Design a CNN architecture to classify arrhythmias in ECG data with the following requirements:

Input: 2-second ECG segments sampled at 200 Hz.

Output: Multi-class classification (e.g., normal, atrial fibrillation, ventricular tachycardia). (4)

i.Describe the layer configuration, including the number of convolutional layers, filters, and activation functions.

ii. Justify how each layer contributes to detecting arrhythmia patterns in ECG segments.

5B) You are tasked with designing a CNN model to analyze ECG data for detecting arrhythmias. You can choose between two architectures: (3)

Option A: Three convolutional layers with filters of increasing size (32, 64, 128) and max pooling after each layer.

Option B: Five convolutional layers with a constant filter size of 64 and max pooling after every second layer.

i.Select the most appropriate option from above and justify your choice based on the ECG data characteristics and computational efficiency.

ii. Explain how the chosen architecture impacts feature extraction and model performance.

5C) A bio-signal processing task requires classifying EEG data into various sleep stages. You have two ANN architectures to choose from:

Option A: A Feedforward Neural Network (FNN) with two hidden layers.

Option B: A Recurrent Neural Network (RNN) with Long Short-Term Memory (LSTM) layers. (3)

i.Choose the architecture most suitable for EEG sleep stage classification and explain why.

ii. Describe how the chosen network handles temporal dependencies in EEG data compared to the alternative option.

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